CSE445
Spring 2025

15/01/2025

Introductory Class

Class Information

- Course Name: MACHINE LEARNING
- Course Code: CSE445
- Credit Hours: 3
- Pre-requisites: MAT361, CSE373
- Class time: 1:00pm–2:30pm
- Classroom: SAC 206
- Office: SAC 1044A

Instructor Information

Experience:

- Assistant Professor, North South University, 09/2022 till Now
- Assistant Professor, Minnesota State University, USA
- Investor & Developer, HIVE blockchain based DAO, USA
- Postdoctoral fellow, New Mexico State University, USA
- PhD (Computer Engineering), New Mexico State University, USA
- MSc (Computer Engineering), Oklahoma State University, USA
- Rollout Engineer, Banglalink
- Lecturer, Eastern University, Dhaka
- BSc (Electrical & Electronic Engineering), BUET

Area of Research: Advanced Computer Architecture and HPC, Embedded Systems and IoT, Robot Dexterity Intelligence, Blockchains

Course Summary

- Introduce students to the core concepts employed in machine learning to solve various problems in artificial intelligence.
- Students will learn the theory behind a range of ML systems and put them to practice in tackling problems in natural language processing or computer vision.
- Topics introduced in the course includes linear regression, logistic regression, classification, support vector machines, decision trees, ensemble learning, dimensionality reduction, clustering, and explainable AI.

Course Outcome (CO)

- Upon successful completion of this course, students will be able to,

- CO1: Gain insights into the ML algorithms – (Mid/Final)
- CO2: Pick the best solution to a problem using statistical analysis - (Mid/Final/Project)
- CO3: Be able to implement an ML project (Project)

Textbook

- "Hands-on Machine Learning with Scikit-Learn, Keras, & TensorFlow", Aurelien Geron, 2nd Edition
- Book (Second Edition) posted in Canvas
- GitHub: https://github.com/ageron/handson-ml

Tentative Marks Distribution

- Attendance: 5%
- Class Performance/Impromptu questions/Assignment/Homework: 15%
- Midterm: 25%
- Final Exam: 25%
- Project: 25%
  - Individual Submission: 10% (Literature review, model description, etc.)
  - Presentation: 10% (2/3 times; proposal, model, results)
  - Final Report: 10% (IEEE format, <25% plagiarism, 90%+ Grammarly score, 3,000+ words)
  - F grade for INCOMPLETE or BAD report

Common Rules

- Questions from alumni: https://docs.google.com/document/d/1FJbrDgz9LOsDcRf53ZTm-0beNcgD7F2ACTmtoxz9dHc/edit?usp=sharing
- Group Project: 4-5 members
- At least 5 ML models (including TWO ensemble)
- Hyperparameter tuning
- Feature Engineering/scaling
- Outlier detection
- Explainable AI
- CSE498R opportunity

Mean, Median, Mode

- Arithmetic mean (average) of a data set is found by adding all numbers in the data set and then dividing by the number of values in the set.
- Mean: Can be skewed by outliers or extreme values
- Harmonic Mean (HM) is defined as the reciprocal of the average of the reciprocals of the data values
- The median is the middle value when a data set is ordered from least to greatest.
- The mode is the number that occurs most often in a data set

Conditional probability

- Conditional probability is a measure of the probability of an event occurring, given that another event (by assumption, presumption, assertion or evidence) has already occurred

Thought Experiment

Thought experiment - performs an intentional, structured process of intellectual deliberation in order to speculate, about potential consequents for an outcome (or consequent). Imagine you have a system that can take a side view picture of an object placed on it and can also measure the weight of the object.

Task: You are to write a program that will identify the object (which is either an apple or an orange) from its image and weight value.

How would you approach it?

Attributes/Features

The measurable features in this thought experiment are:

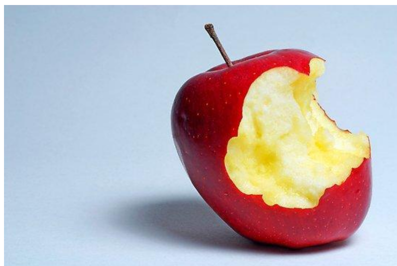
- Image: Color, Shape
- Weight

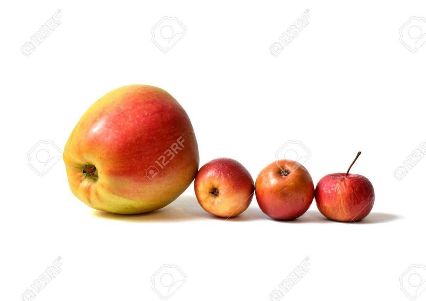
The task is to design a classifier that can classify the type of fruit placed on this machine. Any idea how you can approach it?

Real World

Different shape, color, size, weight, etc.

Large Variations exist

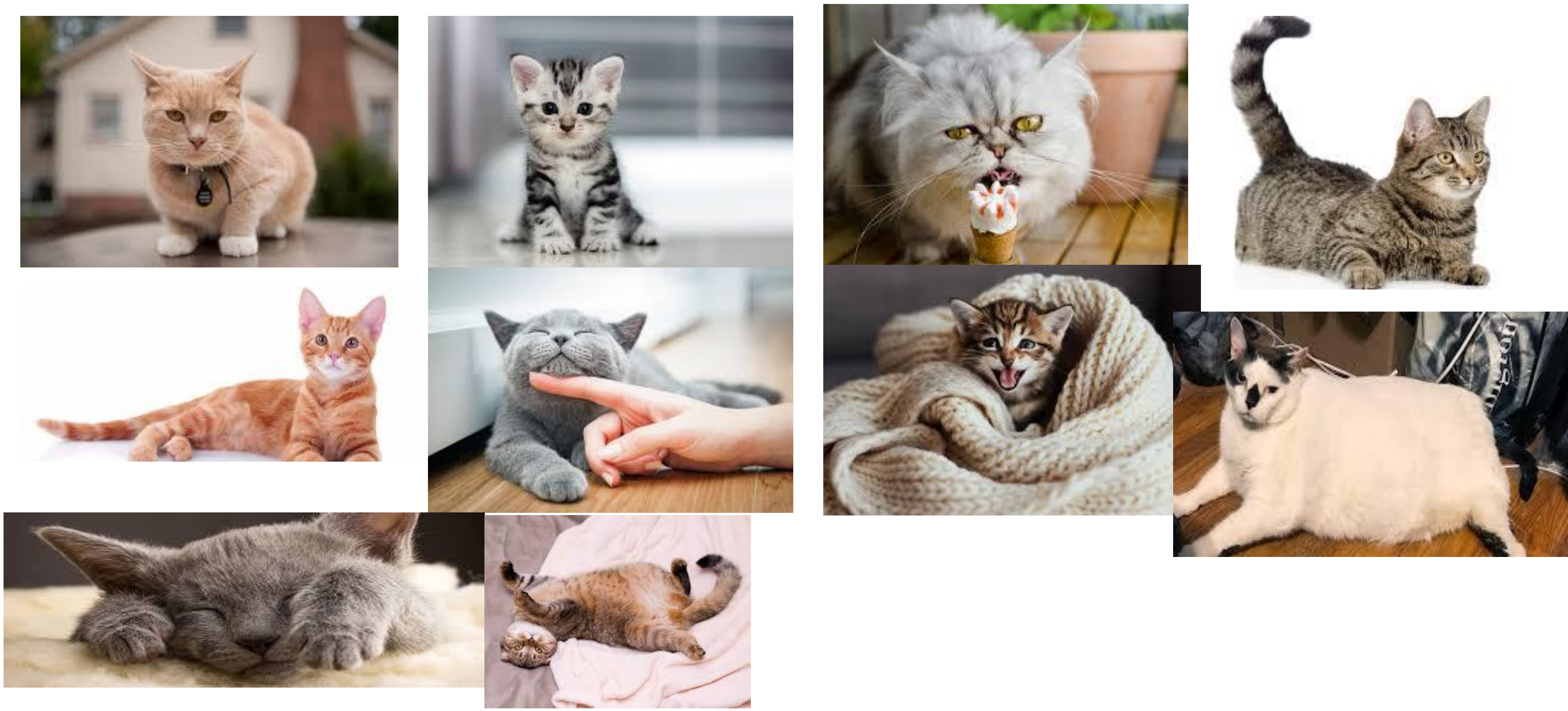




Traditional algorithm, rule-based program fails

Recognizing a cat from a picture

Different background, color, positions, etc.



Real World Data

Noisy

Illumination: lighting effects

Occlusion: surrounded by other objects

Camera angle

Exists a wide range of variation (in terms of features such as color, size, weight, etc..)

Real World Data (cont…)

Consequently, it is impossible to write a traditional rule based programming to recognize a cat or differentiate between fruits (such as orange and apple)

A shift in the paradigm is required

- Data driven Computing (Machine Learning): new field of computational analysis which uses provided data to directly produce predictive outcomes
- We use data to train a machine to do a number of things
- Machine learns distinguishing features/attributes from data
- Data is at the core of ML
- Without data, you will not be able to run a ML project.